CHEM 1301 Driller 9-1

Chapter 9 only — 10 exam-style multiple-choice questions

Instructions. Choose the single best answer (a–e) for each question. You may use a periodic table. Calculators are allowed. Answers are rounded to the precision shown. Time yourself: aim for about 20 minutes (roughly 2 minutes per question, the same pace as the full exam).

Answer sheet. Use the bubble answer sheet (`CHEM1301/exam-prep/answer-sheet/answer-sheet.pdf`), rows 1–10.

1. For the balanced equation $2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$, which statement is correct?

a) 2 mol of H_2 reacts with 1 mol of O_2 to form 2 mol of H_2O
b) 2 molecules of H_2 react to form 1 molecule of H_2O
c) 2 g of H_2 reacts with 1 g of O_2 to form 2 g of H_2O
d) The mass of H_2O formed equals the mass of H_2 used
e) 1 mol of H_2 reacts with 2 mol of O_2

2. For $4Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$, which conversion factor converts **moles of O_2** into **moles of Fe_2O_3**?

a) $\dfrac{3 \text{ mol } O_2}{2 \text{ mol } Fe_2O_3}$
b) $\dfrac{2 \text{ mol } Fe_2O_3}{3 \text{ mol } O_2}$
c) $\dfrac{2 \text{ mol } Fe_2O_3}{4 \text{ mol Fe}}$
d) $\dfrac{3 \text{ mol } O_2}{4 \text{ mol Fe}}$
e) $\dfrac{32.00 \text{ g } O_2}{1 \text{ mol } O_2}$

3. Propane burns according to $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$. How many moles of O_2 are needed to burn 2.40 mol of C_3H_8 completely?

a) 0.480 mol
b) 2.40 mol
c) 7.20 mol
d) 9.60 mol
e) 12.0 mol

4. You are given the mass of reactant A and asked for the mass of product B. Which sequence of conversions is correct?

a) grams A → moles B → moles A → grams B
b) grams A → grams B, using the coefficients as a mass ratio
c) grams A → moles A → grams B
d) grams A → moles A → moles B → grams B
e) grams A → moles A → moles B → atoms B

5. Hydrogen peroxide decomposes: $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$. What mass of O_2 forms when 50.0 g of H_2O_2 decomposes completely?

a) 0.735 g
b) 11.8 g
c) 23.5 g
d) 47.0 g
e) 50.0 g

6. Calcium carbide reacts with water: $CaC_2(s) + 2H_2O(l) \rightarrow C_2H_2(g) + Ca(OH)_2(aq)$. How many moles of H_2O are needed to react completely with 32.0 g of CaC_2?

a) 0.250 mol
b) 0.499 mol
c) 0.998 mol
d) 2.00 mol
e) 18.0 mol

7. Phosphorus burns in oxygen according to the **unbalanced** equation $P(s) + O_2(g) \rightarrow P_4O_{10}(s)$. What mass of O_2 is needed to react completely with 12.4 g of P?

a) 0.500 g
b) 8.01 g
c) 10.2 g
d) 12.8 g
e) 16.0 g

8. For $2CO(g) + O_2(g) \rightarrow 2CO_2(g)$, a mixture contains 3.0 mol of CO and 2.0 mol of O_2. Which statement is correct?

a) CO is limiting, and 0.5 mol of O_2 is left over
b) Neither is limiting; both are used up completely
c) O_2 is limiting, because there are fewer moles of O_2 than of CO
d) CO is limiting, and 1.0 mol of O_2 is left over
e) O_2 is limiting, and 1.0 mol of CO is left over

9. Aluminum reacts with chlorine according to the **unbalanced** equation $Al(s) + Cl_2(g) \rightarrow AlCl_3(s)$. If 10.0 g of Al reacts with 35.0 g of Cl_2, what is the theoretical yield of $AlCl_3$?

a) 0.329 g
b) 43.9 g
c) 45.0 g
d) 49.4 g
e) 65.8 g

10. Limestone decomposes when heated: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$. A student heats 25.0 g of $CaCO_3$ and collects 12.4 g of CaO. What is the percent yield?

a) 0.885%
b) 49.6%
c) 56.0%
d) 88.5%
e) 113%